MIL-D-2457H

11 March 1977

SUPERSEDING
MIL-D-2457G
31 October 1969

MILITARY SPECIFICATION

DUSTER, INSECTICIDE, MASS DELOUSING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size of duster used for mass disinfesting of personnel and clothing.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

FF-H-111	-	Hardware, Builders; Shelf and Miscellaneous
QQ-A-250/8	-	Aluminum Alloy 5052, Plate and Sheet 5052
QQ-A-601	-	Aluminum Alloy Sand Castings
QQ-S-698	-	Steel, Sheet and Strip, Low Carbon
QQ-T-191	-	Terne Sheets (Long and Short Terne)
TT-E-529	-	Enamel, Alkyd, Semi-Gloss
TT-P-1757	-	Primer Coating, Zinc Low Moisture Sensitivity
PPP-C-650	-	Crates, Wood, Open and Covered
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-P-291	-	Paperboard, Wrapping and Cushioning
PPP-T-60	-	Tape, Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3740

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MIL-P-116	-	Preservation-Packaging, Method Of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-S-1837	-	Spray Gun, Insecticide Dusting Powder and Container, Powder
MIL-C-4109	-	Coupling Assembly, Low Pressure, Air-Hose, Quick Disconnect
MIL-E-10062	-	Engines: Preparation for Shipment and Storage Of
MIL-M-10578	-	Corrosion Removing and Metal Conditioning Compound (Phosphoric Acid Base)
MIL-P-17802	-	Padlocks and Padlock Sets, Pin Tumbler Mechanism, Key Operated
MIL-C-23827	-	Grease, Aircraft and Instrument, Gear and Actuator Screw
MIL-T-46786	-	Tank, Fuel, Engine General Requirements for

STANDARDS

FEDERAL

FED-STD-595	-	Colors
FED-STD-601	-	Rubber: Sampling and Testing

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-1410	-	Methods for Selection of Industrial Engines for End Item Application
MS 16535	-	Rivet, Tubular, Oval Head
MS 20470	-	Rivet, Solid-Universal Head, Aluminum and Aluminum Alloy
* MS 35785	-	Cocks, Drain, Ground Key, 150 PSI Steam Pressure
MS 35791	-	Handle, Bail, Chest, Surface Mounting, Countersunk Holes
MS 35822	-	Hinge, Butt, Continuous (Piano), 0.040 Thickness, 0.091 Pin Diameter

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DRAWINGS

US ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

5-13-1918 - Duster, Insecticide, Mass Delousing: Assembly
5-13-1919 - Duster, Insecticide, Mass Delousing; Box, Parts, Shipping and Storage
5-13-1920 - Duster, Insecticide, Mass Delousing; Manifold Assembly and Miscellaneous Details

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY OF MECHANICAL ENGINEERS CODE (ASME)

Power Test Code for Displacement Compressors, Vacuum Pumps and Blowers

(Application for copies should be addressed to ASME, 345 East 47th Street, New York, NY 10017.)

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H-28 - Screw-Thread Standards for Federal Services

(Application for copies of Handbook H-28 should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Dept., 1616 P Street, N.W., Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 108 - Steel Bars, Carbon, Cold Finished, Standard Quality
A 519 - Seamless Carbon and Alloy Steel Mechanical Tubing

(Application for copies should be addressed to the Amercian Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Alternate components. Components offered as equivalent to components specified hereinafter and on the reference drawings as a specific manufacturer's part number or equal shall be functionally equal, or of equal or better quality, with the manufacturer's part number identified. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other components, unless such modification or change is specifically approved by the contracting officer. Prior to manufacture of the first article, or if none is required, prior to commencing production, the contractor shall submit for the contracting officer's approval a list identifying each proposed alternate component, together with proof that each listed component complies with requirements specified above. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of first article or production manufacture as applicable and does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.2 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.2).

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3.3 Materials and components. The materials and components shall be as specified herein and on the drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer for insecticide dusters, provided the complete item complies with all provisions of this specification.

3.3.1 Component parts. The duster shall consist of the following components as specified herein:

Quantity per outfit	Nomenclature
1	Compressor Unit, Power Driven Gasoline Engine, Duster (see 3.5.2)
1	Box, Parts, Shipping and Storage, duster (see 3.5.8)
10	Spray Gun, Including Container (see 3.3.17)
10	Container, Powder (see 3.3.17)
10	Hose, 1/4-inch (6.4 mm) ID, 25 Feet (7.62 m)Long, Complete With Couplers (see 3.3.16)
12	Respirator, Air Filtering, M4 (see 3.3.22)

3.3.2 Carbon steel (except tube).

3.3.2.1 Sheet and strip. Unless otherwise specified hereinafter, steel sheet and strip shall conform to commercial quality of QQ-S-698.

* 3.3.2.2 Bar. Unless otherwise specified hereinafter, steel bar shall be grade 1020, any condition, conforming to ASTM A108.

3.3.3 Terne sheet. Terne sheet shall conform to grade A of QQ-T-191.

* 3.3.4 Steel tube. Steel tube shall conform to ASTM A519.

3.3.5 Copper tube. Copper tube used for air lines to the manifold shall conform to type K of WW-T-799.

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3.3.6 Aluminum alloy sheet. Aluminum alloy sheet used in the fabrication of the parts box and deflector plate shall conform to 5052-H22 or 5052-H32 of QQ-A-250/8.

3.3.7 Aluminum alloy casting. Aluminum alloy casting used in the fabrication of the manifold shall conform to 356-T6 of QQ-A-601.

* 3.3.8 Hinges, continuous, aluminum alloy. Continuous hinges used on the parts box shall be not less than 30 inches (762 mm) long and conform to MS 35822-1A.

3.3.9 Handles, chest. Chest handles used on the parts box shall conform to MS 35791-2.

3.3.10 Hasp, safety hinge. Safety hinge hasp used on the parts box shall conform to type F1401C of FF-H-111.

* 3.3.11 Padlock, 1-1/8 inch (28.6 mm) with chain. The 1-1/8 inch (28.6 mm) padlock shall be brass, with a safety chain, 4 inches (102 mm) (plus or minus one link) long, conforming to grade III, style A, class 1, type A of MIL-P-17802.

* 3.3.12 Cock, drain, 3/8 inch (9.5 mm). The 3/8 inch (9.5 mm) brass drain cock used on the manifold shall conform to MS 35785-7.

3.3.13 Rivets.

* 3.3.13.1 Rivets, universal head, 1/8 inch (3.2 mm) diameter aluminum. Universal head aluminum rivets, 1/8 inch (3.2 mm) in diameter, shall conform to the requirements of MS 20470.

3.3.13.2 Alternate rivets - tubular oval head. Tubular, oval head, 1/8 inch (3.2 mm) diameter, aluminum rivets, may be used in the fabrication of the storage box. Rivets shall conform to the requirements of MS 16535.

3.3.14 Bolts, studs, nuts, and washers. Bolts, studs, nuts, and washers used for mounting components, but not specified shall be the sizes and types recommended by the contractor and shall conform to current commercial standards. Threads shall be class 2 fit and shall conform to Handbook H-28. Steel bolts, studs, nuts, and washers shall be zinc or cadmium plated.

* 3.3.15 Coupling assemblies, hose, air. Air hose coupling assemblies shall be of the quick-disconnect type for 1/4 inch (6.4 mm) hose size conforming to MIL-C-4109. The coupling halves shall be type I, class A, styles 1 and 3, and type II, class A, style 3.

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* 3.3.16 Hose, air. The rubber hose shall be commercial grade, single braid, designed for use with compressed air. The hose shall be 1/4 inch (6.4 mm) I.D. with a tube made from Buna-N rubber and a cover made from neoprene rubber. The color of the cover shall be black. The minimum burst pressure shall be 800 psi (5520 kPa) and the ultimate elongation for tube shall be not less than 250 percent and for the cover 350 percent when tested as specified in 4.4.1.1 (see 6.5).

3.3.17 Spray guns. Spray guns, complete with plastic containers, shall conform to MIL-S-1837.

3.3.18 Grease, bearing. Bearing grease shall conform to the requirements of MIL-G-23827.

* 3.3.19 Primer. Primer for all metal surfaces shall conform to TT-P-1757.

3.3.20 Enamel. Enamel shall be semigloss conforming to TT-E-529. The color shall match color No. 24087 of FED-STD-595.

* 3.3.21 Bolt, spring. The spring bolt shall be 1/2 inch (12.7 mm) square by 6 inches (152 mm) long nominal, with a 6 inches (152 mm) long by 1-3/4 inches (44.5 mm) wide, 0.071 inch (1.80 mm) thick plate without a striker. The spring bolt shall be equipped with a heavy spring to hold the bolt in position. The bolt and plate shall be zinc or cadmium plated.

3.3.22 Government furnished equipment. The M4 air filtering respirators shall be Government furnished and shall conform to the requirements of MIL-R-3308 (see 6.2).

* 3.4 Design. The duster shall consist of an air compressor driven by a gasoline engine and mounted on a common base inclosed within a metal tubular frame. The compressor and engine shall be connected by either a direct drive coupling or timing belt drive. The compressor shall have a manifold mounted on top and shall be equipped with ten quick-operating air hose coupling assemblies, a pressure gage, a drain cock, and a safety relief valve (see 3.3.1). An aluminum parts box, detachable from the metal frame, shall be furnished for storing air hoses, spray guns, spare plastic containers, dust respirators, tools and accessories (see 3.5.9). The net weight of the compressor unit with the gasoline engine shall not exceed 195 pounds (88.5 kg).

3.5 Construction. The construction of the duster shall conform to the requirements specified herein and on drawings referenced in section 2. The finished duster shall be capable of operating without visible signs of leaks, breakdown, shock, vibration, or wear, when tested as specified in 4.5.2, 4.5.3, and 4.5.4, as applicable. When subjected to endurance test specified in 4.5.4, there shall be no significant change in the performance characteristics recorded as specified therein.

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* 3.5.1 Engine, gasoline. The gasoline engine shall be 4-cycle, capable of producing not less than 2.6 net continuous brake horsepower at 2200 revolutions per minute (rpm), radio suppressed, conforming to group I class II of MIL-STD-1410 and the engine exhaust valves shall be equipped with positive type rotators and exhaust valves and seat inserts shall be faced with, or cast from, stellite (see 6.3) or materials of equivalent analysis. All serviceable components of the engine such as air cleaner, magneto, spark plug, gasoline strainer and carburetor shall be easily accessible for service in the engine compartment without removal of the engine, or any other components of the duster. The following specific accessories are considered essential to the satisfactory performance of the gasoline engine and shall be included with the standard specified accessories and auxiliaries.

3.5.1.1 Engine grounding. The installed gasoline engine shall be securely grounded to the frame base with a grounding strap or with flexible wire not less than size No. 12.

3.5.1.2 Engine governor. The engine shall be governed at the required speed to operate the compressor at its rated capacity. There shall be a hand operated attachment for adjusting to the required speed.

* 3.5.1.3 Fuel tank. The fuel tank shall conform to the requirements of, type I of MIL-T-46786 and shall have a capacity of not less than 1 gallon (3.79 liters) and be fabricated from 22 gage (0.0299 inch (0.760 mm)) terne plate conforming to 3.3.3.

3.5.2 Compressor, air. The compressor shall be equipped with an air pressure gage and safety valve. The discharge air from the compressor shall show no evidence of hydrocarbon when tested as specified in 4.5.4. The hydrocarbon content test specified in 4.5.4 is not required when a diaphragm compressor is supplied. The use of exhaust filters or other similar attachment devices shall not be permitted. The nonworking parts of the compressor shall be heat treated cast aluminum of high tensile strength. The compressor shall be cooled by means of fins, with additional cooling being provided by means of a blower connected to the crankshaft that will discharge air over the fins of the head. The blower wheel shall be steel capable of withstanding shock and vibration during normal use. The crankshaft shall be of one piece construction to provide rigidity and shall be equipped with counter balance weights to minimize vibration. The intake and exhaust valves shall be constructed of corrosion-resisting materials and shall be removable. The inlet to the air compressor shall be provided with a removable dry-type air filter. All bearings of the compressor shall be of the antifriction type, either ball or tapered roller, and shall be prelubricated and sealed with grease specified in 3.3.18. Bearings shall conform to commercial, sealed-for-life type. If a diaphragm type compressor is used, the diaphragms shall be compounded of high temperature synthetic rubber and reinforced with three plies of nylon for maximum strength. The compressor shall not leak when tested in accordance with 4.5.2.

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* 3.5.2.1 Capacity. The air compressor, when driven by the gasoline engine specified in 3.5.1, shall deliver 11 cubic feet of oil-free air per minute (0.0052 m3/s) with manifold pressure of 30 pounds per square inch (207 kPa) gage when tested as specified in 4.5.5.

* 3.5.2.2 Flexible coupling. A flexible coupling shall be provided between the gasoline engine and the air compressor when using a direct drive. The coupling shall be constructed with two steel flanges. The coupling shall be Browning Co. No. FS-4 type I or equal.

3.5.2.3 Timing belt. When a timing belt is used it shall require the minimum distance between the shaft centerlines. The gasoline engine and air compressor shall be facing each other with the centerline of each offset to the required distance specified above. The belt shall be fabricated with tension members, neoprene backing, and neoprene teeth with nylon facing. The belt shall be not less than one inch (25.4 mm) wide with a 1/2 inch (12.7 mm) pitch. Standard mating timing pulleys shall be used to obtain the proper speed reduction.

* 3.5.3 Manifold. The body of the manifold shall be cast of aluminum alloy specified in 3.3.7.The manifold shall not leak when tested as specified in 4.5.1. Tube specified in 3.3.5 shall be used for air lines from the manifold to the compressor.

* 3.5.3.1 Gage, pressure. The pressure gage shall have a polycarbon plastic case, silver gray color, with a clear polycarbon plastic crystal. The polycarbon plastic shall be of the high impact type. The gage shall have a 1/4 inch National Pipe Thread (N.P.T.) male bottom connection. The dial shall be 2 inch (150.8 mm) size constructed from brass with white background and black markings to read from 0 to 160 psi. The bourdon tube used in the gage shall be fabricated from seamless phosphor bronze tube, heat treated and seasoned. The bourdon tube, dial movement, and pointer shall be assembled as a complete unit on the socket, the case merely acting as an inclosure. The gage shall be United States Gauge No. P563, or equal.

* 3.5.3.2 Valve, safety. The safety valve shall be spring loaded, with a manual relief ring, a brass body, and a corrosion-resisting steel seat. The safety valve shall have a 3/8 inch N.P.T. male connection and shall be preset to discharge at 30 psig (207 kPa) when tested as specified in 4.5.4. The safety valve shall have a pressure rating of 150 psig (1034 kPa).

* 3.5.4 Hoses with couplings. Each duster shall be provided with 10 spray hoses, each not less than 25 feet (7.6 m) long, conforming to 3.3.16. Each spray hose shall be fitted with hose couplings specified in 3.3.15. One end of the hose shall have a type II, class A, style 3 half coupling specified in 3.3.15, and the other end shall have a type I, class A, style 3 half coupling attached. Hose clamps shall be Tinnerman's type A3122-10-1 or equal.

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3.5.5 Spray guns and plastic containers. Spray guns and plastic containers shall conform to 3.3.17 and shall be provided in the quantities specified in 3.3.1.

* 3.5.6 Plate, heat deflector. The heat deflector plate shall be approximately 6 by 12 inches (152 by 304 mm), constructed of aluminum alloy specified in 3.3.6. The deflector plate shall be free from sharp corners and rough edges and shall be braced to prevent the plate from vibrating.

* 3.5.7 Framework and mounting. The entire duster shall be mounted within a steel tubular frame. The frame shall be fabricated from steel tube specified in 3.3.4. All components of the frame shall be welded. Mounting of the engine compressor assembly shall be on a steel base with rubber bushings to absorb vibrations. Torsional vibration induced at rated output of the compressor when coupled to the engine shall be minimized by dynamic balancing of the compressor. The base shall be fabricated from steel not less than 14 gage, (0.75 inch (1.91 mm)) conforming to 3.3.2.1. Side edges of the base shall be turned down approximately 7/8-inch (22.2 mm) and notched to fit the frame at the ends. The ends of the base shall be securely welded to the frame.

* 3.5.7.1 Skids. Steel strip, 1/8 by 1-1/2 inches (3.2 by 38.1 mm), conforming to 3.4.1.1, shall be welded to the bottom of the tubular frame to form skids. Skids shall extend up around the radius of the tube approximately 3 inches (76.2 mm) on each end. Ends of the skids shall be rounded.

3.5.8 Parts box. The parts box shall be fabricated from aluminum alloy sheet specified in 3.3.6 and shall be constructed as shown on Drawing 5-13-1919.

3.5.9 Tools and accessories. Each duster shall be complete with the following tools and accessories (see 6.4):

	Quantity (each)
Starter rope, engine	1
Pliers, slip joint, straight nose with cutters, 6 inches (152.4 mm) long	1
Screwdriver, flat type, 3/8 inch (9.5 mm) wide	1

3.6 Welding. Welds shall show no evidence of separation when tested as specified in 4.5.4. Fillet welds shall have equal legs and no overlaps.

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3.7 Riveting. Rivets shall be of the type and size indicated on drawings and shall be of such length as to permit the rivets to be securely and neatly set. Rivets shall be driven to completely fill the holes, with full size heads neatly finished and in full contact with the surface of the riveted members.

3.8 Finish. Primer and enamel shall be applied to the exterior surfaces of the outfit, except for the rubber hoses, and where not desirable for proper operation of the unit such as moving parts, gages, and couplings.

3.8.1 Surface preparation. All metal surfaces to be enameled shall be thoroughly cleaned and conditioned with metal conditioner conforming to MIL-M-10578.

3.8.2 Priming. All surfaces prepared as specified in 3.8.1 shall be coated with primer specified in 3.3.19.

3.8.3 Enameling. After primer is dried, all primed surfaces shall be coated with enamel specified in 3.3.20. There shall be no heavy runs, blisters, peeling, or imbedded foreign matter in the enamel finish.

3.9 Marking for identification. The item covered by this specification shall be marked for identification in accordance with end item marking requirements of MIL-STD-130.

3.10 Workmanship. The finished duster shall be clean and all sharp burrs, slivers, or splinters shall be removed. All surfaces to be soldered shall be thoroughly cleaned prior to soldering. Soldering joints shall be smooth, sound, and clean. All bolts, nuts, screws, and studs shall be drawn tight. Finish shall be smooth and free from blisters, peeling, chipping, or areas of no film. There shall be no indication of rust, corrision, scale, or dirt under, or in, finish. Welding shall be free from slag inclusion, undercuts, incomplete sections, cracks, fractures, or incomplete fusion. Scale and flux deposits shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article shall be examined for defects in table I, dimensions and weight specified, and shall be tested in accordance with 4.5.2, and 4.5.4. The presence of any visual defect, any dimension or weight not within the specified requirements, or failure to pass the tests shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1 components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.1.1 Testing of hoses. Testing of the hoses shall be performed on specimens of hose. The lot shall be all hose manufactured under the same process offered for inspection at one time. The sample unit shall be one specimen cut from the hose lot. One sample unit shall be subjected to each test. The sample units shall be tested for burst strength and ultimate elongation in accordance with Method 10011 and Method 4121 respectively of FED-STD-601. Any nonconformance to the requirements of 3.3.16 shall constitute failure of the tests, and rejection of the lot.

4.4.2 In-process inspection. Examination shall be made of the following operations to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be made to the items affected and to the operation.

4.4.2.1 Visual examination of operations.

a. Rivets are correct size and properly driven (see 3.7).
b. Metal surfaces cleaned and conditioned prior to painting as specified (see 3.8.1).
c. Metal surfaces primed as specified (see 3.8.2).

4.4.2.2 In-process testing.

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4.4.2.2.1 Manifold test. Each manifold, after fabrication, and prior to application of finish shall be tested in accordance with 4.5.1. Any defect shall be cause for rejection.

4.4.2.2.2 Compressor test. One compressor from each lot manufactured under the same process shall be tested in accordance with 4.5.5. Failure to pass the test shall be cause for rejection of the lot of compressors manufactured under the same process.

4.4.3 End item inspection. A lot shall consist of all dusters offered for inspection at one time. The sample unit for these inspections shall be a completely fabricated duster.

4.4.3.1 Visual examination. The completely assembled end item shall be examined for defects classified in table I. The inspection level shall be II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Type of finish not as specified		X
	Finish applied where not desirable for proper operation of the unit	X	
	Finish blistered, peeled, chipped, or area of no film		X
	Not finished where required	X	
	Finish not continuous, smooth and adherent (i.e., crack, runs, drips, lap, creep, orange peel, streaks, wrinkles, or flaky)		X
	Indication of rust, corrosion, or scale under finish		X
	Separation of color, or finish rough i.e., lint, dirt, grease, or other foreign matter imbedded in finish		X
Design	Design characteristic not in accordance with the specified requirements	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (general) (applicable to components and assemblies)	Part missing or not type specified	X	
	Fractured, split, punctured, dented, bowed, or malformed	X	
	Part misplaced, loose, or not in proper alignment	X	
	Sharp burr, sliver or projection	X	
	Operation omitted or not properly performed		X
	Functioning component that requires abnormal force to operate	X	
	Not connected or joined as specified	X	
	Adjustable assembly that cannot be adjusted	X	
	Adjustable assembly not properly adjusted to perform the function intended		X
	Component not readily accessible for servicing where specified		X
Welding	Not type specified, missing, incomplete, burn holes, cracked, fractured, or not fused	X	
	Slag inclusion, undercut, not smooth or uniform		X
	Scale or flux deposits not removed		X
	Poor surface appearance of weld as from craters, piling up of deposited metal, evidence that weld has not been ground or filled where required, or grinding or filling not neatly accomplished		X
Bolts, nuts, screws, studs and other types of fasteners	Missing, broken, stripped, fractured	X	
	Loose		X

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Soldering (when applicable)	Missing, not adherent or is incomplete	X	
	Not clean (flux or flux residue not removed), not smooth (surfaces not finished neatly) or pin holes in solder		X
Lubrication (when applicable)	Not lubricated where required	X	
Gages, valves couplings	Not properly mounted	X	
Marking identification	Missing, incomplete, not legible, not specified type or size, not in proper location, not affixed in the prescribed manner, plate not fabricated of the specified material, not stenciled, stamped, embossed, printed, lithographed, et cetera, as applicable		X
Instruction plate	Missing, incomplete, not legible, not in proper location	X	

4.4.3.2 Dimensional and weight examination. Inspection shall be made of the end item for compliance with dimensions, and weight specified. Any noncompliance with specified requirement shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.4.3.3 Testing of the end item. Each production unit shall be tested as specified in 4.5.2 and 4.5.3. Failure to pass these tests shall be cause for rejection of the unit.

4.4.4 Packaging inspection. An examination shall be made to determine that preservation-packaging, packing, and marking requirements of section 5 are complied with. Defects shall be as listed in table II. The sample unit shall be one shipping container fully prepared for delivery except it shall not be sealed. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

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TABLE II. Inspection of preservation-packaging

Examine	Defect
Markings	Missing; incorrect; illegible; of improper size, location, sequence, or method of application
Preservation	Preservative improperly applied or missing
Materials	Any nonconforming component; component missing or damaged
Workmanship	Inadequate application of components such as components of parts box not adequately cushioned and immobilized Bulging or distorted container

4.4.4.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-C-650, examination for defects in closure and strapping shall be in accordance with the appendix of that specification.

4.5 Tests.

4.5.1 Manifold. After complete fabrication of the manifold, but prior to application of finish or preservative and after plugging of all openings, the manifold shall be filled with compressed air at a pressure of 150 psig (1471 kPa) and fully submerged in water to determine compliance with 3.5.3.

4.5.2 Connections. The compressor shall be operated and permitted to build up an air pressure of 30 psig (207 kPa). All connections, including air lines, shall be tested for air leaks by the application of water containing a 20 percent solution of soap to determine compliance with the requirements specified in 3.5 and 3.5.2. The test may be performed during the operating test specified in 4.5.3.

4.5.3 Operating test. The complete unit, without a load of insecticide, shall be given an operational test of 2 hours duration. Any nonconformance of mechanical equipment to function as specified in 3.5 shall constitute failure of this test.

4.5.4 Endurance test. The compressor, with its gasoline engine, shall be subjected to 200 hours of continual operation at full rated capacity to determine compliance with the requirements specified in 3.5, 3.5.2 and 3.6. The 200 hours of operation may be in increments of not less than 20 hours per day, except for interruptions required for normal servicing of the equipment. Should a major breakdown occur, the endurance test shall be rerun. A major breakdown is defined as a breakdown requiring more than 1 hour during operation for any repair including all contingencies. During the test, the following data shall be recorded at 1 hour intervals and any nonconformance to the requirements of 3.5.1, 3.5.2 and 3.6 shall constitute failure of the test:

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(a) Back pressure of gasoline engine exhaust system (measured 2 inches (50.8 mm) from the exhaust manifold (flange)).
(b) Engine speed.
(c) Compressor crank shaft speed.
(d) Engine-oil sump temperature.
(e) Engine cyclinder-head temperature.
(f) Compressor discharge pressure.
(g) Compressor discharge temperature.
(h) Compressor capacity (corrected).
(i) Hydrocarbon content of discharge air.

Failure of the compressor to operate continually at rated capacity for 200 hours without a major repair; failure of the compressor to discharge oil-free air; failure of the safety valve to operate as specified in 3.5.3.2; or back pressure on the exhaust system in excess of 8 inches (203 mm) of water shall constitute failure of this test.

* 4.5.5 Capacity test. The compressor component shall be driven by a direct-coupled dynamometer or a calibrated electric motor. The compressor activity shall be determined by the ASME Low Pressure Nozzle Method of the Power Test Code for Displacement Compressors, Vacuum Pumps, and Blowers utilizing the following formula:

$$Ga = \frac{2.557CDn2\ T1}{Pa} \frac{(Pb\ p)}{(T2)}\ 1/2$$

GA = Volumetric rate of flow cubic feet per minute (c.f.m.)
T1 = Absolute temperature at compressor intake Fahrenheit (F)

NOTE: Ambient temperature shall be determined at engine crankshaft height 3 to 5 feet from the compressor.

Pa = Absolute pressure at compressor intake, pounds per square inch (psi).
C = Nozzle coefficient of discharge.
Dn = Nozzle throat diameter (inches).
Pb = Barometric pressure (inches Hg).
T2 = Absolute temperature, upstream of the nozzle (F).
p = Differential pressure across the nozzle (inches water).

The compressor speed required to produce rated capacity shall be recorded. A capacity of less than 11 c.f.m. (0.0052 m3/s) at 30 psig (207 kPa), or horsepower required in excess of 2.6 shall constitute failure of the test. Other test methods by the industry are acceptable, providing results conform to the above specified test.

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5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Preservation-packaging shall be in accordance with the method and material requirements of MIL-P-116 as hereinafter specified. Where tape is specified, it shall conform to type III or IV, class 1 of PPP-T-60.

5.1.1.1 Cleaning and drying. All exposed, uncoated ferrous metal surfaces of each duster shall be cleaned according to process C-1 and thoroughly dried.

5.1.1.2 Preservation.

5.1.1.2.1 Application of preservative. Surfaces, cleaned and dried as specified and that do not contact other surfaces in operation, shall be coated with type P-1 preservative. Other cleaned surfaces shall be coated with type P-6 preservative. Surfaces coated with type P-6 preservative which might be removed by contact with packaging and packing materials shall be covered or wrapped with greaseproof barrier material conforming to type II, grade A of MIL-B-121, and the wrap secured with tape.

* 5.1.1.2.2 Gasoline engine and air compressor. The gasoline engine shall be preserved in accordance with level A requirements for type II, Method I of MIL-E-10062, except that the air compressor shall not be preserved, but all openings on the compressor shall be sealed with tape. The air pressure gage on the air compressor shall be protected with single-faced corrugated fiberboard secured with tape.

* 5.1.1.2.3 Components. Both ends of each spray hose shall be sealed and thoroughly covered with tape. The 10 discharge tubes shall be disassembled from the spray guns, wrapped together with single-faced corrugated fiberboard, and secured with tape or twine. The two openings into each spray gun shall be sealed with tape. Each respirator shall be preserved method IC-1 or placed in an individual polyethylene bag. The polyethylene film shall be minimum 0.002-inch (.05 mm) thick. The mouth end of the bag, when prepared for closure, shall extend a minimum of 3 inches (76.2 mm) above the packaged item. Closure of the bag shall be effected by heat sealing.

5.1.1.3 Packaging. The 10 spray hoses, 10 spray guns, 10 plastic containers, equipment manual, discharge tubes for spray guns and respirators, tools and accessories, preserved as specified in 5.1.1.2, shall be packaged in the appropriate compartments of parts box conforming to Drawing 5-13-1919. The spray guns, plastic containers and tools shall be cushioned with paperboard conforming to type I, style 1 or 2 of PPP-P-291, or cellulosic cushioning

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material conforming to type I or II, class A of PPP-C-843 to prevent movement during transit. Empty spaces between the packaged components and the cover of parts box shall be fitted with rolls, pads, cells, or die-cut sheets made of the same type paperboard or cellulosic cushioning material specified. The cover of the parts box shall then be secured with the padlock and the padlock key wired to the hasp with minimum 0.055-inch (1.4 mm) diameter annealed wire. Handles on the box shall be secured to the box to prevent movement, and the opening between cover and body of box shall be sealed with tape.

5.1.2 Commercial. Dusters shall be preserved and packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A packing. Each duster, preserved and packaged as specified in 5.1, with the parts box attached in its position on top of the outfit, shall be packed in a snug-fitting crate conforming to type I, style A of PPP-C-650. Wood blocking shall be nailed to the bottom of the crate under the baseplate of the duster to reinforce the baseplate. Wood blocks shall also be placed between the gasoline engine and compressor and the baseplate to remove tension and compression from the rubber mountings. The duster shall be anchored and protected with a full-height shroud in accordance with the appendix of PPP-C-650.

5.2.2 Level B packing. Each duster, preserved and packaged as specified in 5.1, shall be packed as specified in 5.2.1, except that a shroud shall not be required.

5.2.3 Commercial packing. Each duster, preserved and packaged as specified in 5.1, shall be packed in a container in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification as applicable.

5.3 Marking.

5.3.1 Military Packaging. In addition to any special marking required by the contract or order, submethod 1C-1 packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial Packaging. Commercial packaging marking shall reflect as a minimum, the item name, national stock number, quantity, unit of issue, contract or order number, and appropriate address markings. The markings shall be applied with a waterproof material, shall be legible, and shall provide a definite contract.

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6. NOTES

6.1 Intended use. The duster covered by this specification is intended for use in the field for disinfesting personnel and clothing without removing the clothing from the body.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is not required (see 3.2).
(c) Source and delivery of Government furnished equipment (see 3.3.22).
(d) Selection of the applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

6.3 Stellite is a product of the Haynes Stellite Division of Union Carbide Co., Kokomo, IN 46901 (see 3.5.1).

6.4 Tools and accessories. The tools and accessories listed in 3.5.9 may be further identified by the Federal Stock Number (NSN) listed below:

Pliers - NSN 5120-00-223-7396
Screwdriver - NSN 5120-00-278-1280

* 6.5 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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